Qi Zhou



qizhou1729.github.io
https://github.com/qizhou1729
zhouqi1729@sjtu.edu.cn

Educational Background

Shanghai Jiao Tong University Sept. 2018 – June 2022

Bachelor — Major: Mathematics and Applied Mathematics (Wen-Tsun Wu's class)
Minor: Computer Science and Technology

Shanghai Jiao Tong University Sept. 2022 – Now

Ph.D. Candidate (Combined Master and Doctoral) — Computational Mathematics (Advisor: Zhenli Xu)

CCM, Flatiron Institute, Simons Foundation May 2025 – Aug. 2025

Intern Research Associate (Mentor: Shidong Jiang)

Department of Mathematics, KTH Royal Institute of Technology Jun. 2026 – Dec. 2026

Visiting Ph.D. Student (Mentor: Anna-Karin Tornberg)

Academic Grade

- GPA: 4.00/4.00
- Rank **1** over **79** students.
- Graduate courses (Courses in bold received a full mark):
  - *General Fundamental Courses:* English for Academic Purposes (A), Scientific Writing, Integrity and Ethics (A), Specialized English (A+).
  - *Core Courses:* **Analysis (A+)**, **Scientific Computing (A+)**, **Partial Differential Equations (A+)**, Measure Theory and Probability (A+).
  - *Elective Courses:* **Algebraic Combinatorics (A+)**, Integrability and Bifurcation Theory of Dynamical Systems (A+), **Advanced Computation Methods (A+)**, Applied Mathematical Methods (A+), Variational Methods and Degree Theory (A+).
  - *Frontier Courses:* Additive Combinatorics in Number Theory (A+), High-Performance Computation for Differential Equations (A+).

Fund and Project

NSFC Young Scientist Fund (Ph.D. Program), PI Jan. 2026–Dec. 2027
Youth Sci-Tech Talent Cultivation Project of CAST (Ph.D. Program) Jan. 2026–Dec. 2027

Research Interest and Publications

I am currently interested in fast algorithms of molecular dynamics (MD) simulation, especially scalable algorithms for general interaction kernels. Also, I consider the design of specialized MD hardwares and supercomputers and the application of new algorithms, expected to approach the SOTA. I am also interested in other fast algorithms such as convolution equations and non-local interactions. Publications are listed as follows:

1. (Alphabetic order) J. Liang, Z. Xu and **Q. Zhou**, Random batch sum-of-Gaussians method for molecular dynamics simulations of particle systems, *SIAM J. Sci. Comput.*, 45 (2023), B591-B617.
2. (Alphabetic order) J. Liang, Z. Xu and **Q. Zhou**, Error estimate of the u-series method in molecular dynamics simulations, *Appl. Comput. Harmon. Anal.* 77 (2025) 101.
3. W. Gao, T. Zhao, Y. Guo, J. Liang, H. Liu, M. Luo, Z. Luo, W. Qin, Y. Wang, **Q. Zhou**, S. Jin and Z. Xu, RBMD: A molecular dynamics package enabling to simulate 10 million all-atom particles in a single graphics processing unit, *Comm. Comp. Phys.*, 39 (1) (2026), 296-322.
4. (Alphabetic order) Z. Xu, Y. Zhao and **Q. Zhou**, Variance-reduced random batch Langevin dynamics, *J. Chem. Phys.*, 161(2024), 244110.

5. (Alphabetic order) X. Gao, S. Jiang, J. Liang, Z. Xu and **Q. Zhou**, A fast spectral sum-of-Gaussians method for electrostatic summation in quasi-2D systems, *Numer. Math.*, 158 (2026), 533-585.
6. (Alphabetic order) Y. Lin, Z. Xu, Y. Zhang and **Q. Zhou**, Weighted balanced truncation method for approximating kernel functions by exponentials, *Phys. Rev. E*, 112 (2025), 015302.
7. X. Gao[#], **Q. Zhou**[#], Z. Gan[*] and J. Liang[*], Accurate error estimates and optimal parameter selection in Ewald summation for dielectrically confined Coulomb systems, *J. Chem. Theory Comput.*, 21 (12) (2025), 5890-5904. **Special issue "Developments of Theoretical and Computational Chemistry Methods in Asia"**.
8. Y. Tu, H. Tian, L. Shi, **Q. Zhou**, Q. Zhang and W. Mao, SchulzNN: A neural network-based matrix inversion solver inspired by Schulz iteration, submitted.
9. **Q. Zhou**[#], T. Wu[#], J. Liu, Q. Sun, H. Xie and Z. Xu, Sum-of-Gaussians tensor neural networks for high-dimensional Schrödinger equation, arXiv:2508.10454.
10. W. Gao, **Q. Zhou**[*], Q. Zhang and Z. Xu[*], Symmetry-preserving random batch Ewald method for constant-potential simulation of electrochemical systems, *J. Comput. Phys.*, 555 (2026), 114789.
11. (Alphabetic order) X. Gao, S. Jiang, J. Liang and **Q. Zhou**, An $O(\log N)$ Monte Carlo method for periodic Coulomb systems, *SIAM J. Sci. Comput.*, in press, arXiv:2601.09288.
12. (Alphabetic order) Z. Jiang, J. Liang and **Q. Zhou**, Random batch sum-of-Gaussians method for molecular dynamics simulation of particle systems in the NPT ensemble, arXiv:2602.23582.
13. T. Wu[#], **Q. Zhou**[#], H. Zheng, H. Xie and Z. Xu, Spectral convergence of sum-of-Gaussians tensor neural networks for many-electron Schrödinger equation, *J. Chem. Phys.*, 164 (24) (2026), 244103. **Special collection "Machine Learning for Electronic Structure"**.
14. (Alphabetic order) S. Jiang, D. Wang and **Q. Zhou**, A fast sum-of-Gaussians algorithm for the high-dimensional fractional Fokker-Planck equation, arXiv:2606.28184.
15. **Q. Zhou**, Y. Guo, J. Zhong, S.-H. Bo, T. Zhao and Z. Xu, RBMD 2.0: Random batch molecular dynamics package for large-scale simulations on multi-GPU architectures, arXiv:2609.02694.

Patent

1. Co-inventor, "A Random batch sum-of-Gaussians method for molecular dynamics simulations of particle systems", Patent number: ZL 2022 1 0449795.9
2. Co-inventor, "A molecular dynamics simulation system", Publication number: CN120183519A
3. Co-inventor, "A sum-of-Gaussians tensor neural network method for solving many-body Schrödinger equation", Publication number: CN120975253A
4. Co-inventor, "A parallel molecular dynamics simulation system with high scalability", Publication number: CN122154366A

Award and Scholarships

Zhiyuan honorary scholarship	Apr. 2019
Honorable mentioned in 2019 MCM/ICM	Apr. 2019
The third prize (provincial level) in 2019 CUMCM	Oct. 2019
The first prize (national level) of the 11-th Chinese mathematics competition, qualified for final	Nov. 2019
Finalist in 2020 MCM/ICM	Apr. 2020
Distinguished prize in 2nd Alibaba international mathematics competition	July 2020
Chia-Chiao Lin bronze medal in 11-th Yau's mathematics competition	Oct. 2020
Distinguished prize in 11-th Yau's mathematics competition (Analysis)	Nov. 2020
The second prize (provincial level) in 2020 CUMCM	Nov. 2020
The first prize (national level) of the 12-th Chinese mathematics competition	Dec. 2020
The second prize of the 11-th final Chinese mathematics competitions	Apr. 2021
Outstanding winner, COMAP scholarship and SIAM award in 2021 MCM/ICM	Apr. 2021
Chia-Chiao Lin silver medal in 12-th Yau's mathematics competition	May 2021
Baosteel outstanding student scholarship	Sept. 2021
The 5-th Sensetime scholarship	Nov. 2021
The third prize (collegiate level) of 3rd SJTU life science innovation competition	Dec. 2021
The second scholarship of outstanding undergraduate in SJTU	Dec. 2021
Distinguished prize in 13-th Yau's mathematics competition (Analysis)	Aug. 2022

Distinguished prize in 13-th Yau's mathematics competition (Applied Math)	Aug. 2022
First-class graduate scholarship	Dec. 2022
National scholarship for graduate (M.S.)	Sept. 2023
SJTU merit student	Oct. 2023
First-class graduate scholarship	Dec. 2023
Bonditech spark scholarship	May 2024
Excellent oral presentation in the 8-th CSIAM Student Forum	Oct. 2024
Huatai securities technology scholarship	Nov. 2024
Best oral presentation in the 6-th greater bay area symposium on soft and living matter	Jan. 2025
National scholarship for graduate (Ph.D.)	Sept. 2025
SJTU excellent teaching assistant	Aug. 2026
National scholarship for graduate (Ph.D.)	Sept. 2026